INCH-POUND

MIL-I-5914E
15 OCTOBER 1993
SUPERSEDING
MIL-I-5914D
5 DECEMBER 1979

MILITARY SPECIFICATION

INDICATOR, RATE OF FLOW, FUEL, TYPE EFU-24/A AND TYPE EFU-25/A

This specification is approved for use by the Department of Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of sealed, single remote indicating, synchro, fuel flow rate indicators, range 100 to 2,000 pounds per hour (pph).

1.2 Classification. Indicators shall be furnished in the following type designations and dial markings, as specified (see 6.2):

Type	EFU-24/A	Flourescent dial and pointer markings. (Replaces Type A-16)
Type	EFU-25/A	White dial and pointer markings.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : Oklahoma City Air Logistics Center/TICLA, Tinker AFB, OK 73145-3037 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620
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SPECIFICATIONS

FEDERAL

PPP-B-1672	Boxes, Shipping, Reusable With Cushioning

MILITARY

MIL-P-116	Preservation, Method Of
MIL-T-5913	Transmitter, Rate Of Flow, Synchro, Type J-5
MIL-I-6599	Indicator, Rate of Flow, Type A-19
MIL-I-7057	Indicators, Synchro, Aircraft, General Specification For

STANDARDS

FEDERAL

FED-STD-101	Preservation, Packaging And Packing Material: Test Procedures
FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U S Military Property
MIL-STD-2073-1	DOD Materiel Procedures For Development And Application Of Packaging Requirements
MS33585	Pointer, Dial Standard Design Of Aircraft Instrument
MS33639	Case, Instrument, Clamp Mounted Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are issues of the documents cited in the solicitation (see 6.2).

American Society For Testing and Materials (ASTM)
D910 Gasolines, Aviations

(Applications for copies should be address to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. The documents also my be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable lows and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Indicators furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.2 and 6.4).

3.2 General. The requirements of MIL-I-7057 are applicable as requirements of this specification. Additional requirements shall be as specified herein. In the event the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.3 Design and construction.

3.3.1 Outline dimensions. The outline dimensions of the indicator shall conform to MS33639, for the 2-inch size and the length"L" shall be ± 0.125 inches.

3.3.2 Case. The case shall be hermetically sealed. A separate nonhermetically sealed cap may be used on the back of the case for the purpose of installing the electrical receptacle.

3.3.3 Indicator dial. The indicator dial shall conform to Figure 1.

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3.3.4 Pointer. The pointer shall conform to MS33585-8, except that the length of the pointer shall be such that the tip will overlap the shortest graduation by a distance equal to 1/3 to 2/3 of the length of the shortest graduation.

3.3.4.1 Pointer finish. The shaded portion of the pointer shall be finished with lusterless white, color No. 37875 of FED-STD-595 flouresent-luminescent material in accordance AF P/N 898894.

3.3.5 Electrical receptacle. The electrical receptacle shall conform to MS33678-14S-2P or MS33678C-14S-2P if a nonhermetically sealed cap is used on the back of the instrument case.

3.4 Calibration. Each indicator shall be calibrated and the dial graduated in accordance with Figure 1. The calibration of fuel flow versus angular degrees shall be linear within ± Ø degrees, 15 minutes of arc.

3.4.1 Calibrated adjusted. Each indicator shall be calibrated and adjusted for maximum accuracy at the rate of flow specified in Table I as the "setting point".

TABLE I. Fuel Flow Rate Indicator

Flow rate, pph	Tolerance, pph
100	5.Ø
200	5.Ø
400	5.Ø
600	5.Ø
800	5.Ø
1,000	5.Ø
1,200	5.Ø
1,400	5.Ø
1,600	5.Ø
1,800	5.Ø
2,000	5.Ø

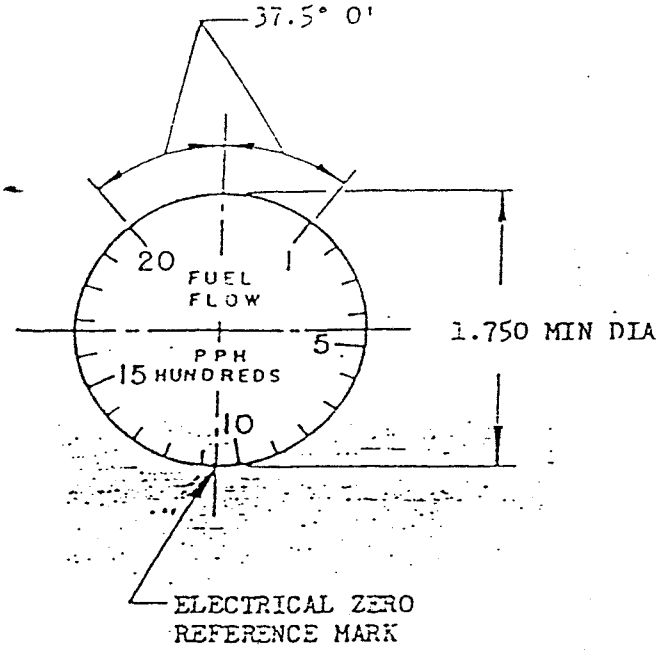
3.4.2 Remote station. Unless otherwise specified, the fuel flow rate indicator shall be designed to indicate at a remote station, the rate of fuel flow as transmitted by a type J-5 transmitter in accordance with MIL-T-5913. The indicator shall increase in value when a voltage sequence of ADC is applied at the indicator connector pins.

3.5 Weight. The weight of the complete indicator shall not exceed Ø.7Ø pound.

3.6 Markings.

3.6.1 Dial Markings. The following markings shall be finished in lusterless

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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES:

ANGLES ±0° 15'

FIGURE 1. Dial.

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white, color No. 37875 to FED-STD-595 flourescent-luminescent material, or in accordance with AF P/N 898894. The dimensions of the marking shall be:

	Height or length, (inch) + Ø.Ø16)	Width of line or graduation, (inch) + Ø.ØØ5
Scale numerals 1, 5, 1Ø, 15 and 2Ø	Ø.156	Ø.Ø31
Graduations opposite numerals 1, 5, 1Ø, 15, and 2Ø	Ø.156	Ø.Ø31
1ØØ-pound graduations	Ø.Ø94	Ø.Ø2Ø
Lettering "PPH"	Ø.Ø62	Ø.Ø16
Lettering "HUNDREDS"	Ø.Ø62	Ø.Ø16
Lettering "FUEL FLOW"	Ø.Ø94	Ø.Ø2Ø

3.6.1.1 Color. The marking "Type EFU-24/A" or "Type EFU-25/A", as applicable, shall be permanently and legibly marked on the dial. These markings, and all other markings not otherwise specified, shall be finished in lusterless black, color No. 37Ø38 of FED-STD-595. The height of the markings shall be Ø.Ø47 ± Ø.Ø16 inch.

3.7 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-13Ø. The following special characteristics shall be added:

26V, 4ØØ Cycles

3.8 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with

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all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirement, however, this does not authorize submission of know defective material,either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of Inspection. Qualification inspection, quality conformance inspection and test conditions shall be in accordance with MIL-I-7057 and as specified herein.

4.3 Quality conformance inspection. The quality conformance inspections shall consist of the following individual and sampling tests. Performance of the indicator for scale error and friction shall be checked against Tables I and II of this specification.

4.3.1 Individual tests. Each indicator submitted for acceptance under contract shall be subjected to the individual tests specified in MIL-I-6599.

4.3.2 Sampling tests. Sampling tests shall be as specified in accordance with MIL-I-7057. Failure of any sample to meet requirements therein, shall cause rejection of the lot.

4.3.3 Test Samples shall consist of three indicators representative of production equipment. Samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4 Inspection and test. Test methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification

TABLE II. Fuel Flow Indicator

Test	Tolerance, pph
Position error at 1,050 pph	2
Friction:	
Room temperature	10
High temperature	10
Low temperature	20
Salt fog	4
Vibration:	
Pointer oscillation	5
Pointer variation	5
Scale error:	
Low temperature	7
High temperature	5
Salt fog	4

4.5 Retention of qualification. Each vender listed on the Qualified

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Products List (QPL) will be required to certify their product on DD form 1718 every two years.

5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C" as specified (see 6.2).

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in quantity unit pack (QUP) of one each. Each item will be provided a preservation method IC-1 IAW MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116 C-1 process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073-1 to provide a method IC-1.

5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply sources to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073-1.

5.2 Packing. Packing shall be level "A", "B", or "C", as specified (see 6.2).

5.2.1 Container-cushioning. Intermediate container requirements shall conform to PPP-B-1672, weather-resistant, unless otherwise specified by the contracting activity. Intermediate container shall be large enough to allow for application of sufficient cushioning material, between the unit container and intermediate container. Cushioning material shall be of sufficient density and thickness to insure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Fiberboard containers do not meet Level "A" container criteria.

5.2.3 Level B. Item will be preserved as specified in 5.1 and shall be packed in exterior container conforming to PPP-B-1672, weather-resistant, unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with the protection required. Closure shall be in accordance with appropriate PPP-B-1672 procedures, as specified by contractor, special packaging instructions (SPI), and contracting activity.

5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical damage during direct domestic shipment

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from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073-1.

5.3 Markings. Unit, intermediate, and exterior container shall be marked IAW special markings required by the contractor, special packaging instructions (SPI) and MIL-STD-129.

5.4 Inspection and Test. Test of methods of preservation shall be accomplished in accordance with section 4 of MIL-P-116 to insure compliance with section 5 of this specification. Packaging tests shall be conducted IAW rough handling as specified in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but not mandatory.)

6.1 Intended use. The fuel flow indicator, is conjunction with the J-5 transmitter, is intended to indicate the rate of flow of fuel in pph being consumed by an aircraft engine.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Data requirements.
(c) Dial and pointer finish desired (see 3.3.4.1 and 3.6).
(d) Applicable level of packaging and packing (see 5.1 and 5.2).
(e) Type designation (see 1.2).
(f) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Qualification . With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contacts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is OC-ALC/TICLA Tinker AFB, OK 73145-3037 and information pertaining to qualification of products may be obtained from that activity.

6.4 Subject term (key word) listing.
Fuel flow rate indicators
Range 100 to 2000 pph

6.5 Charges from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the

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extensiveness of the changes.

Custodian:
AIR FORCE -99
Navy -AS

Reviewer Activity
AIR FORCE -11

Preparing Activity:
AIR FORCE -71

Project Number:
6620-0505

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-5914E	93/10/15

3. DOCUMENT TITLE

INDICATOR, RATE OF FLOW, FUEL TYPE EFU-24/A AND EFU-25/A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
OC-ALC/TICLA n		336-5960
c. ADDRESS (Include Zip Code) TINKER AFB OK 73145-3037	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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